

PROBLEM SET 3

ECON301—ECONOMETRICS I

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1. Let $\{W_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables with $E(W_i) = 0$ and $\text{Var}(W_i) = \sigma_W^2$, show that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n W_i \rightarrow_d N(0, \sigma_W^2)$$

2. Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of random variables such that $E(X_n) \rightarrow c$ and $\text{Var}(X_n) \rightarrow 0$ as $n \rightarrow \infty$. Show that then $X_n \rightarrow_P c$.
3. **Chebychev's second inequality.** Let X be a random variable with $E(X) = \mu < \infty$ and $\text{Var}(X) = \sigma^2$ with $0 < \sigma^2 < \infty$. Then for any real number $k > 0$,

$$\Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Prove this result.

4. Suppose $\bar{X}$ is the mean of 100 observations from a population with mean μ and variance $\sigma^2 = 9$. Find limits between which $\bar{X} - \mu$ will lie with probability at least 0.90 using:
 - i. The second Chebychev's inequality from item 3.
 - ii. The Central Limit Theorem.

Comment on your results. Which limits are tighter?

5. Let $\{Y_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables with Bernoulli probability mass functions with unknown parameter p . Let $\hat{p}_n = \frac{1}{n} \sum_{i=1}^n Y_i$.
 - i. Show that $\hat{p}_n$ is a consistent estimator of p .
 - ii. Find and justify the large sample distribution of

$$\frac{\hat{p}_n - p}{\sqrt{\frac{p(1-p)}{n}}}$$

- iii. Find and justify a consistent estimator of $\text{Var}(Y_1)$.
- iv. Find and justify the large sample distribution of

$$\frac{\hat{p}_n - p}{\sqrt{\frac{\hat{p}_n(1-\hat{p}_n)}{n}}}$$

- v. A sample of size 127 is such that $\sum_{i=1}^{127} Y_i = 39$. Find a 99% confidence interval for p .
- vi. Suppose we are interested in the *odds ratio*, $\gamma = p/(1-p)$, which is the probability of success over the probability of failure. Propose a consistent estimator of γ and show that it is indeed consistent. Call your proposed estimator $\hat{\gamma}_n$.
- vii. Consider the following theorem:

Theorem (Delta Method). If $\sqrt{n}(T_n - \tau) \rightarrow_d N(0, \psi^2)$ and $U_n = h(T_n)$ where h is continuously differentiable at τ , then

$$\sqrt{n}(U_n - h(\tau)) \rightarrow_d N(0, [h'(\tau)]^2 \psi^2).$$

Equivalently, if $T_n \sim AN(\tau, \psi^2/n)$ and $U_n = h(T_n)$ is continuously differentiable at τ , then

$$U_n \sim AN\left(h(\tau), \frac{[h'(\tau)]^2 \psi^2}{n}\right).$$

Use the Delta Method to show that $\sqrt{n}(\hat{\gamma}_n - \gamma) \rightarrow_d N(0, p/(1-p)^3)$.

- viii. Propose a consistent estimator of $p/(1-p)^3$ and show that it is indeed consistent. Call your proposed estimator s_n^2 .
 - ix. Give the details of building an asymptotic pivot to make inference on γ .
 - x. Construct an asymptotic $(1-\alpha)\%$ confidence interval for γ .
6. The lifetime of a certain computer chip that a company manufactures is characterized by the population distribution

$$f(z; \theta) = \frac{1}{\theta} e^{-z/\theta} I_{(0, \infty)}(z),$$

where z is measured in thousand of hours. Let $X_1, \dots, X_n$ represent n i.i.d. random variables with the density $f(z; \theta)$. An outcome of the random variable

$$Y_n = \frac{1}{n} \left(1 + \sum_{i=1}^n X_i \right)$$

will be used to provide an estimate of θ .

- i. Is it true that $Y_n \rightarrow_{L_2} \theta$? Is it true that $Y_n \rightarrow_P \theta$?
 - ii. Define an asymptotic distribution for Y_n .
 - iii. Suppose that $n = 100$, and $\theta = 10$. use the asymptotic distribution you defined in (b) to approximate the probability that $Y_n \geq 15$.
7. Let $X_1, \dots, X_n$ be i.i.d. random variables having continuous uniform distributions of the form $f(z) = I_{[0,1]}(z)$.
- i. Define an asymptotic distribution for $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$.
 - ii. Using your result from (a), argue that

$$\left(\sum_{i=1}^{12} X_i \right) - 6 \approx Z \sim N(0, 1).$$